



THE FEDERAL UNIVERSITY OF TECHNOLOGY, AKURE
SCHOOL OF ENGINEERING AND ENGINEERING TECHNOLOGY
DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING
1ST SEMESTER B.ENG. EXAMINATION 2023/2024 SESSION
EEE 311 ANALOG COMMUNICATION PRINCIPLES

INSTRUCTIONS: Answer any four (4) questions. Carefully read all questions, provide equations and diagrams where needed. Course Unit: 2 units.

Time Allowed: 2 hrs.

Question 1 (15 marks)

- (a). In an FM system, a 5 kHz modulating frequency signal modulates 100 MHz carrier wave so that frequency deviation is 50 kHz. Find the highest and lowest frequency attained by the FM signal. (4 marks)
- (b). In an FM system, a 7 kHz modulating frequency signal modulates 107.6 MHz carrier wave so that frequency deviation is 50 kHz. Find the carrier swing in the FM signal and the modulation index. (3 marks)
- (c) A rule of bandwidth for FM signal is sometimes used as

$$BW = (2m_f + 1)f_m$$

where m_f is the modulation index. Find the fraction of the signal power that is included in that frequency band when m_f is 1 and when m_f is 10. (8 marks)

Question 2 (15 marks)

- (a) The modulating signal in an FM wave is 500 Hz with amplitude 3.2 V and frequency deviation of 6.4 kHz. If the audio frequency voltage is now increased to 8.4 V, determine the new frequency deviation and modulation index. If the audio frequency voltage is raised to 20 V while the audio frequency is dropped to 200 Hz, determine the frequency deviation and the modulation index. (6 marks)
- (b) In a DSB-FC AM, show that for 100% modulation, each of the side band contains one-sixth of the total power. (6 marks)
- (c) Show that for a single-tone AM, the maximum transmission efficiency is 33.33%. (3 marks)

Question 3 (15 marks)

- (a) Use appropriate expressions and diagrams to show how a square-law modulator can be used to generate AM wave. (8 marks)
- (b) Use sketch to show that the information content in an AM wave is contained in the amplitude variation. (2 marks)
- (c) Determine the percentage power saving when the carrier and one of the sidebands are suppressed in an AM wave modulated to a depth of;
- (i) 50% (5 marks)

Question 4 (15 marks)

- (a) Let $m_1(t)$ and $m_2(t)$ be two message signals, and let $x_{c1}(t)$ and $x_{c2}(t)$ be the modulated signals corresponding to $m_1(t)$ and $m_2(t)$, respectively. Prove that if the modulation is DSB(AM), then $m_1(t)+m_2(t)$ will produce a modulated signal equal to $x_{c1}(t)+x_{c2}(t)$. (5 marks)
- (b) The antenna current of an AM transmitter is 10 A when it is modulated to a depth of 30% by an audio signal. It increases to 12 A when another signal modulates the carrier signal. What will be the modulation index due to the second signal? (10 marks)

Question 5 (15 marks)

- (a) State the reason why the bandwidth of FM wave is difficult to estimate. (2 marks)
- (b) Determine the permissible range in the maximum modulation index for a commercial FM which has 30 Hz to 15 kHz modulating frequencies. (5 marks)
- (c) A single-tone FM wave is represented by the following equation $v = 40 \cos[50000t + 6 \sin 3000t]$ volts. What type of FM is this? Also, estimate
- (i) The carrier and the modulating frequencies
 - (ii) The maximum deviation
 - (iii) The bandwidth of the FM signal using Carson's rule. (8 marks)